Install and Run C in Visual Studio Code

Step-by-step manual for Windows 10 and Windows 11 | GCC (MinGW-w64 from WinLibs) + Visual Studio Code

By the end of this guide, you will have installed the GCC compiler, configured Windows PATH, installed VS Code extensions and run a C program.

1. Download MinGW-w64 from WinLibs

1. Visit https://winlibs.com/.

2. Scroll to **Download** and choose a recent GCC + MinGW-w64 build for **Win64 (x86_64)**. A UCRT or MSVCRT build with POSIX threading is suitable.

3. Download the ZIP archive, right-click it and choose **Extract All**.

4. Find the **mingw64** folder and copy it directly into **C:**.

Check that the compiler exists at the following exact location:

```
C:\mingw64\bin\gcc.exe
```

Tip: Some ZIP archives contain extra enclosing folders. Move the inner mingw64 folder so there is no doubled path, such as C:\mingw64\mingw64\bin.

2. Add MinGW-w64 to the Windows PATH

1. Open Start and search for **Edit the system environment variables**.

2. Click **Environment Variables**.

3. Under **User variables**, select **Path**, then click **Edit**.

4. Click **New** and paste the folder below.

5. Click **OK** in all windows to save.

```
C:\mingw64\bin
```

Open a **new** Command Prompt and verify the installation:

```
gcc --version
gdb --version
```

You should see version details. If Windows says gcc is not recognized, review the PATH entry and open a new terminal.

3. Install Visual Studio Code

1. Go to https://code.visualstudio.com/download.

2. Download the Windows installer and run it.

3. Follow the prompts. If available, enable **Add to PATH** and **Open with Code**.

4. Finish installing, then open Visual Studio Code.

4. Install the required extensions

Inside VS Code, press **Ctrl + Shift + X** to open the Extensions panel. Install both:

C/C++ by Microsoft - language support, IntelliSense and debugging.
C/C++ extension page

Code Runner by Jun Han - provides the Run Code command.
Code Runner extension page

Note: The C/C++ extension does not install GCC. You need the MinGW-w64 compiler from Step 1.

5. Set the C compiler path

1. Press **Ctrl + ,** to open VS Code Settings.

2. Search for **C_Cpp.default.compilerPath**.

3. In **C_Cpp: Default: Compiler Path**, enter:

```
C:\mingw64\bin\gcc.exe
```

This setting tells the C/C++ extension which compiler to use for IntelliSense.

6. Enable Code Runner in the integrated terminal

1. In Settings, search for **code-runner.runInTerminal**.

2. Tick **Code-runner: Run In Terminal**.

3. This is essential for programs that read keyboard input using scanf().

7. Alternative: update settings.json directly

Press **Ctrl + Shift + P**, select **Preferences: Open User Settings (JSON)**, then add these settings inside the existing JSON object:

```
{
  "C_Cpp.default.compilerPath": "C:\\mingw64\\bin\\gcc.exe",
  "code-runner.runInTerminal": true
}
```

Important: If the file contains other settings, do not overwrite them. Separate entries with commas. Backslashes in JSON paths must be doubled.

8. Write your first C program

1. Create a folder named **C_Programs**.

2. In VS Code, choose **File > Open Folder** and select C_Programs.

3. Create a file named **hello.c** and enter:

```
#include <stdio.h>

int main(void) {
    printf("Hello, World!\n");
    return 0;
}
```

Save the file using **Ctrl + S**.

9. Compile and run the program

Option A: Code Runner. With hello.c open, choose **Run Code** (not the separate Run/Debug command), or press **Ctrl + Alt + N**.

Option B: VS Code terminal. Go to **Terminal > New Terminal** and enter these commands in the folder containing hello.c:

```
gcc hello.c -o hello.exe
.\hello.exe
```

Expected program output:

```
Hello, World!
```

10. Verify keyboard input works

Create a file named **input.c**:

```
#include <stdio.h>

int main(void) {
    int age;
    printf("Enter your age: ");
    scanf("%d", &age);
    printf("You are %d years old.\n", age);
    return 0;
}
```

Save and choose **Run Code**. Enter a number when prompted. If the number is read and printed correctly, your terminal configuration works.

11. Troubleshooting

gcc is not recognized: Check PATH contains C:\mingw64\bin and reopen the terminal.

gcc.exe not found: Look inside C:\mingw64\bin and correct your compiler path if necessary.

Cannot type input: Set code-runner.runInTerminal to true and use the Code Runner Run Code command.

Red underlines on stdio.h: Check the C/C++ compilerPath setting and reload VS Code.

Old output after changes: Save your file before running it.

Final checklist

■ GCC installed ■ PATH updated ■ gcc --version works
■ VS Code installed ■ C/C++ extension installed
■ Code Runner installed ■ compilerPath configured
■ Run in Terminal enabled ■ hello.c runs successfully